

Question	Answer	Marks
9(a)	spontaneous emission of (ionising) radiation	B1
	emission (of radiation) from an unstable nucleus	B1
9(b)(i)	$\lambda = \ln 2 / (12.0 \times 60)$	C1
	$= \mathbf{9.63 \times 10^{-4} \text{ s}^{-1}}$	A1
9(b)(ii)	$A = (5.80 \times 10^{14}) \exp(-\ln 2 \times 20.0 / 12.0)$	C1
	$= \mathbf{1.83 \times 10^{14} \text{ Bq}}$	A1
9(b)(iii)	<u>sketch:</u> line starting at (0, 5.80)	B1
	line with negative gradient from $t = 0$ to $t = 20.0$ min with gradient of continuously decreasing magnitude	B1
	line passing through (12, 2.90) <u>and</u> (20, 1.83)	B1
9(c)	<u>any three points from:</u> - radiation is emitted in all directions but only some is towards detector - some radiation is absorbed within the sample - some radiation is absorbed in the air (before reaching detector) - some radiation is absorbed by the window of the detector - some radiation reaches the detector but does not register a count	B3